

UDCT v7.10 - Post-registration

Function sensitivity and persistent local Jeans obstructions

Registered GQUMOND virial and stellar-profile tests in three ultra-faint dwarf galaxies

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Status. Numerical results report under the reviewed v7.10 prospective protocol dated 5 October 2026 [1]. The known v7.09 results motivated that protocol. The new calculations reported here were executed on 8 October 2026. This report preserves the registration and records the existing pre-execution implementation amendment. No new physical law or successful UDCT model is claimed.

Abstract

The registered calculation tests four one-at-a-time function variants and four mass- and projected-radius-preserving softened profiles in a specified implementation of Milgrom's GQUMOND example structure. All numerical self-tests and the v7.09 reproduction gate pass. The complete execution comprises 849 registered case-grid evaluations plus six triggered truncation follow-ups. The registered outcomes are **G2, J1 and K2**. The $f_{1/2}$ screening variant at a length of 10 pc produces negative global virial work for both Hydrus I profiles on all three grids, while all 30 relevant QUMOND controls remain positive. Every tested function variant retains at least one negative conditional isotropic Jeans-pressure case in the Hydrus I cusp. At 10 pc, negative pressure persists for all four softened profiles, including $\epsilon/r_h = 0.3$. The baseline 1 pc pressure minimum becomes an interior minimum on the separate smaller-inner-limit diagnostic. These results identify function and profile sensitivity under frozen assumptions; they do not establish a universal MOND obstruction, an observational rejection, or a verdict on dynamical UDCT.

Keywords: GQUMOND; virial work; conditional Jeans pressure; screening; Hydrus I; preregistration; numerical reproducibility.

1. Frozen theory and execution protocol

“UDCT” identifies the research series. The theory tested is the specified GQUMOND implementation inherited from v7.09, with Milgrom [4] as its direct theoretical precedent. The earlier UDCT v7.06 action is a different candidate; its historical failures are not rerun here.

$$P = f(u) Q(Z/f), \quad Z = |\nabla\psi|^2/a_0^2,$$
$$u = |\nabla\psi|^2/(\ell_0^2 \psi_{,ij} \psi_{,ij}),$$
$$\nabla^2\phi = (a_0^2/2) [\partial_i(\partial P/\partial\psi_{,i}) - \partial_i\partial_j(\partial P/\partial\psi_{,ij})].$$

Both the leading-gradient and action-derived Hessian sources are retained. The campaign convention is $Q'(Y^2) = v(Y)$, $Q(0) = 0$. The acceleration scale is $a_0 = 1.082401 \times 10^{-10} \text{ m s}^{-2}$. The six fixed screening lengths are 0.03, 0.1, 0.3, 1, 3 and 10 pc.

Variant	Screening f	Q derivative v(Y)
BASE	$u/(1+u)$	$v(1 + 1/Y)$
A-f1/2	$u^{1/2}/(1+u^{1/2})$	Baseline v
A-f2	$u^2/(1+u^2)$	Baseline v
A-Qs	Baseline f	$1/2 + v(1/4 + 1/Y)$
A-Qd	Baseline f	$v[(1 + v(1 + 4/Y^2))/2]$

Inputs are Leo IV, Pegasus III and Hydrus I, fixed stellar mass-to-light ratio 2, Plummer and tapered-cusp profiles, and the inherited visible Galactic disk plus $8.62 \times 10^9 M_\odot$ point bulge. The external field is a local second-order Taylor approximation. No dark halo, LMC field or time evolution is included.

Part B replaces the cusp factor $1/r$ by $1/\sqrt{(r^2 + \epsilon^2)}$, re-solves the scale length and normalizes stellar mass to preserve the same projected half-light radius and total mass. The four ϵ/r_h values are 0.01, 0.03, 0.1 and 0.3. Here r_h always means the unsoftened baseline half-mass radius.

Grid	Radial	Polar x azimuth	Disk R x ϕ x z
Smoke	180	16 x 24	40 x 24 x 8
Default	360	24 x 36	120 x 64 x 12
Refined	720	48 x 72	240 x 128 x 24

There are 282 common case specifications per grid: BASE 42, A 156, B 84. Part C adds three refined-only evaluations: 849 in total. The primary global set is 216 A/B GQUMOND cases and 30 relevant controls. Six triggered 5 r_h follow-up evaluations are additional.

2. Numerical gates and reporting audit

The original runner matches the hash printed in the reviewed PDF. The existing amended runner is dated 5 October 2026 and applies the reviewed strict-negative, exact-zero and invalid-data rules. Comparison of function syntax trees confirms unchanged physical kernels, profile geometry, meshes, case enumeration and integral routines. Changes affect dependency verification, validation, classification, synthetic tests and reporting. The amendment is disclosed before interpretation; no new code correction was made during this execution.

Gate	Measured check	Result
R: baseline reproduction	42 default virials: maximum relative difference 1.20943×10^{-7} (limit 10^{-5}). Three refined pressure minima: 1.10874×10^{-6} (limit 10^{-3}).	PASS
S: constitutive derivatives	All five laws pass independent finite-difference checks. Largest gradient error 6.63×10^{-10} ; largest Hessian-coefficient error 5.86×10^{-5} (limit 10^{-4}).	PASS
Geometry	$\epsilon = 0$ geometry difference 2.2×10^{-16} ; softened enclosed mass absolute error 4.4×10^{-7} ; projected fraction error 4.8×10^{-6} (limits 10^{-5}).	PASS
L: limiting behavior	QUMOND reduction, vanishing extra Hessian tensor, and v high-/low-acceleration checks pass.	PASS
Decision logic	Synthetic zero, missing, NaN/infinite, grid-disagreement and gate-R cases pass.	PASS

These checks establish the specified numerical identities at test states, not physical realism or accuracy throughout every possible parameter domain. The independent outcome audit agrees with the runner: G2/J1/K2. There are no missing or nonfinite primary records, virial sign disagreements, or unresolved default/refined pressure flags in the registered set.

Common 282 cases	Default vs smoke	Default vs refined
Maximum relative virial difference	1.281×10^{-6}	7.672×10^{-8}
Maximum absolute pressure-minimum difference ($\text{km}^2 \text{ s}^{-2}$)	0.002505	0.000613

The maximum refined relative disagreement between the direct-force and integration-by-parts virial values is 1.468×10^{-4} . The integration-by-parts result retains the boundary term and remains primary. Pressure differences are reported separately from virial differences: stable signs do not establish equally precise depths or band endpoints.

3. Global outcome G2 and triggered follow-ups

The inward-positive angular-mean radial force defines $V = \int \rho_*(r) r^3 \langle g_r \rangle dr$, normalized by the positive stellar-Newtonian V_N . A negative QUMOND detection requires strict negativity on smoke, default and refined grids and a strictly positive paired control. An exact zero is a boundary case.

G2: two of the 216 primary A/B QUMOND cases have negative global work on all three grids. The remaining 214 are positive. All 30 relevant QUMOND controls are strictly positive. Both negative cases use A-f1/2, Hydrus I, $\ell_0 = 10$ pc.

Profile	Smoke V/V_N	Default	Refined	Paired control (refined)
Plummer	-7.009045	-7.009036	-7.009036	3.994097
Tapered cusp	-5.557278	-5.557278	-5.557278	3.556883

The full-boundary refined minimum $\langle g_r \rangle / g_*$ over 0.1 - $10 r_h$ is approximately -12.255 for Plummer and -10.926 for the cusp. Each crosses the inherited outward-force warning threshold of -10 on every grid. The prescribed $5 r_h$ follow-up is therefore triggered for these two cases only.

$5 r_h$ follow-up	Smoke V/V_N	Default	Refined
Plummer	-8.100850	-8.100853	-8.100853
Tapered cusp	-6.104542	-6.104543	-6.104543

The follow-up remains negative and does not replace the full-boundary registered outcome. Both versions are archived. This sensitivity check does not authorize a revised physical boundary condition or a claim of a self-consistent finite stellar system.

Refined diagnostic	Plummer	Tapered cusp
Minimum σ_r^2 ($\text{km}^2 \text{s}^{-2}$)	-1.85949	-1.69635
Radius of pressure minimum (r/r_h)	0.2478	0.08132
Sampled negative-pressure band (r/r_h)	0.002 - 1.922	0.002 - 1.731
Sampled outward-force band (r/r_h)	0.002 - 2.260	0.002 - 2.128

Bands are sampled endpoints, not fitted roots. In the inherited code, formal aperture RMS is $\sqrt{\text{max}(\text{aperture second moment}, 0)}$. Its archived value is zero for these cases; this clipped diagnostic is not a physical zero dispersion. All negative-pressure aperture outputs are formal.

4. Local function outcome J1

The conditional spherical isotropic Jeans reduction uses $p_r(r) = \rho_* \sigma_r^2 = \int_r^{R_{out}} \rho_*(s) \langle g_r(s) \rangle ds$ with $p_r(R_{out}) = 0$. A negative σ_r^2 is incompatible with a physical velocity variance under this reduction. Negative-pressure status requires strict negativity on both default and refined grids; smoke is supplementary.

J1: all four A variants retain negative-pressure cases in the unsoftened Hydrus I cusp. The screening exponent 1/2 extends the affected sampled lengths down to 0.3 pc. The other three variants retain the baseline pattern at 1, 3 and 10 pc.

Law	Affected ℓ_0 (pc)	Refined minima at those lengths (km ² s ⁻²)
BASE	1, 3, 10	-0.113062, -0.435382, -1.571442
A-f1/2	0.3, 1, 3, 10	-0.005860, -0.103724, -0.399024, -1.696351
A-f2	1, 3, 10	-0.139003, -0.693307, -2.424807
A-Qs	1, 3, 10	-0.110385, -0.442211, -1.607479
A-Qd	1, 3, 10	-0.118574, -0.450287, -1.616537

A-f1/2 additionally gives negative local pressure in the Hydrus I Plummer profile at 3 and 10 pc. The refined minima are -0.250497 and -1.859490 km² s⁻², respectively. At 3 pc its global work is still positive ($V/V_N = 0.680761$). This explicitly illustrates the distinction between a positive global sign and admissible local pressure.

No negative-pressure cases appear in Leo IV or Pegasus III on the prescribed meshes. No relevant QUMOND control has negative pressure. Across the 282 common specifications, there are 23 negative-pressure cases: three unsoftened BASE, 15 function variants and five softened profiles. All default/refined signs agree. The three additional C cases are also negative, bringing the refined count including diagnostics to 26.

Boundary-limited minima require restricted interpretation. Examples include the 1 pc cusp minima for BASE, A-f2, A-Qs and A-Qd and the 0.3 pc A-f1/2 cusp minimum. Their negative signs count under registration, while their depth and inner extent are not fully resolved on the prescribed inner boundary. No sign tolerance was introduced after seeing the results.

5. Profile outcome K2 and inner-mesh diagnostic

Part B changes finite-density profiles while preserving stellar mass and projected half-light radius. It also changes the density and Newtonian field across a finite radial range. It is a sensitivity experiment, not an isolation of the single point $r = 0$.

ϵ/r_h	$\ell_0 = 1$ pc	3 pc	10 pc
0.01	No negative flag	-0.260936 *	-1.577512
0.03	No negative flag	No negative flag	-1.241709 *
0.1	No negative flag	No negative flag	-0.369577 *
0.3	No negative flag	No negative flag	-0.118706 *

Table values are refined minimum σ_r^2 in $\text{km}^2 \text{s}^{-2}$; flags agree with default. * Minimum at the inner mesh node. “No negative flag” means nonnegative on the prescribed meshes, not a demonstrated physical equilibrium.

K2: negative pressure remains at $\epsilon/r_h = 0.3$ for 10 pc. At 1 pc, every tested softened profile has no negative flag. At 3 pc, only $\epsilon/r_h = 0.01$ remains negative, giving an adjacent sampled transition bracket (0.01, 0.03]. At 10 pc, all four sampled softenings remain negative; there is no sampled disappearance bracket. These four-point flags are monotone, but no continuous monotonicity or exact critical ϵ is inferred.

All 72 Part B GQUMOND virials and 12 softened-profile controls remain positive on all grids. No other object develops a pressure obstruction under the sampled softened profiles.

Part C: a boundary-limited minimum becomes interior

ℓ_0 (pc)	Baseline min σ_r^2	C min σ_r^2	C r/r_h	C boundary limited?
1	-0.113062	-0.133977	0.001186	No
3	-0.435382	-0.435395	0.003790	No
10	-1.571442	-1.571505	0.013104	No

Pressure units are $\text{km}^2 \text{s}^{-2}$. Part C lowers only the refined inner radius from $0.002 r_h$ to $0.0005 r_h$. The baseline 1 pc minimum is resolved into the interior of this diagnostic mesh and becomes more negative. The 3 and 10 pc minima change little. C does not replace the outcome meshes or alter $G/J/K$.

6. Interpretation and reproducibility record

v7.09 showed positive global work for one frozen f/Q law. v7.10 shows that this favorable sign does not persist across every registered function choice: f1/2 at 10 pc changes the sign in Hydrus I. Nevertheless, baseline, f2 and the two Q variants retain positive global signs throughout their tested unsoftened cases. The scientific conclusion is sensitivity to the chosen constitutive function and scale under fixed assumptions.

J1 establishes persistence of the Hydrus I cusp pressure obstruction across the four one-at-a-time choices. K2 shows that the tested finite-density modification does not remove it at 10 pc even at $\epsilon/\rho = 0.3$. The milder lengths respond differently. Neither result establishes that a central singularity is the unique cause, that every regular core fails, or that all GQUMOND choices are excluded.

No likelihood, observational significance or matched-aperture dispersion fit is computed. The prescribed Galactic field omits the LMC, nonaxisymmetric structure and time dependence. Spherical isotropic closure omits rotation, anisotropy and a self-consistent distribution function. No coupled matter-field evolution, bidirectional projection, dynamical back-reaction or lensing is tested. The older UDCT candidate is neither rescued nor newly rejected by these calculations.

Execution, hashes and implementation record

Execution: 8 October 2026, Auckland; full-run start 04:34:57.757663 UTC (17:34:57 Auckland). Python 3.12.14; NumPy 2.3.5; SciPy 1.17.0; Linux 6.18.44 x86_64, glibc 2.39. BASE reproduction was checked before interpretation of variants. The full run regenerated all BASE cases.

Original reviewed runner

ac0c3359ecc7f2d04bdcc81f8922c13c
b3d28c56cb09162d6a4216622d54929d

Executed pre-execution amended runner

a4d1a07ebcaa06348be3022d0f1f6ca4
63347e00194f7fde59f2bd9dbcd521b7

Frozen v7.06 dependency

a64440e1350ae8bf10eb3188ec34829b
6d586abf275de5cdc841a50ad3478284

Runner basename: UDCT_v7_10_Preregistration_GQUMOND_
Sensitivity_and_Hydrus_I_Central_Profile_run(2).py
Self-test command: python3 "RUNNER" --selftest
Full registered command: python3 "RUNNER"
The command-line array with the exact executed path is preserved in results.json.

Archive: UDCT_v7_10_Postregistration_Execution_2026-10-08.zip contains raw case-keyed summaries, all grids, warnings and follow-ups, independent_audit.json, both runner versions, dependency and README. Deviation log: existing 5 October reporting amendment used; no new physical or numerical change and no outcome-driven retuning. Original files retain their deposit-time "not yet run" statements. The PDF reports their subsequent execution.

Author-directed numerical execution, auditing and drafting were assisted by OpenAI ChatGPT/Codex. Original protocol design used Anthropic Claude; earlier research used Gemini and Grok. This report is an independent numerical research record, not peer review. No v7.10 deposit identifier or GitHub repository URL is asserted here.

Appendix A. Function-variant virial matrix

Refined V/V_N , Part A. All entries have the same strict sign on the three grids. Rounded numbers summarize the case-keyed raw archive; a printed equality after rounding is not an exact computational equality.

Law / object / profile	.03 pc	.1 pc	.3 pc	1 pc	3 pc	10 pc
A-f1/2 / Leo IV / Plummer	14.003129	13.976360	13.899926	13.632999	12.875338	10.279390
A-f1/2 / Leo IV / cusp	13.885291	13.859285	13.785193	13.528122	12.809626	10.427377
A-f1/2 / Pegasus III / Plummer	19.358298	19.322488	19.220278	18.863789	17.855691	14.441544
A-f1/2 / Pegasus III / cusp	19.202565	19.166802	19.065026	18.713076	17.737376	14.559179
A-f1/2 / Hydrus I / Plummer	3.960911	3.883480	3.662282	2.888463	0.680761	-7.009036
A-f1/2 / Hydrus I / cusp	3.528823	3.463370	3.276522	2.624347	0.775162	-5.557278
A-f2 / Leo IV / Plummer	14.014605	14.014605	14.014605	14.014605	14.014599	14.013960
A-f2 / Leo IV / cusp	13.896449	13.896449	13.896440	13.895926	13.886263	13.724237
A-f2 / Pegasus III / Plummer	19.373652	19.373652	19.373652	19.373652	19.373622	19.370025
A-f2 / Pegasus III / cusp	19.217916	19.217915	19.217894	19.217092	19.196082	18.842782
A-f2 / Hydrus I / Plummer	3.994097	3.994097	3.994097	3.994097	3.994097	3.994086
A-f2 / Hydrus I / cusp	3.556883	3.556883	3.556882	3.556778	3.554179	3.488777

Appendix A. Function-variant matrix (continued)

Law / object / profile	.03 pc	.1 pc	.3 pc	1 pc	3 pc	10 pc
A-Qs / Leo IV / Plummer	14.488395	14.488381	14.488261	14.486890	14.474839	14.338122
A-Qs / Leo IV / cusp	14.369918	14.369857	14.369328	14.363597	14.319645	13.953013
A-Qs / Pegasus III / Plummer	19.854734	19.854705	19.854450	19.851548	19.826047	19.537987
A-Qs / Pegasus III / cusp	19.698697	19.698582	19.697575	19.686628	19.602819	18.941560
A-Qs / Hydrus I / Plummer	4.409680	4.409671	4.409594	4.408715	4.400989	4.313106
A-Qs / Hydrus I / cusp	3.953460	3.953416	3.953025	3.948634	3.911676	3.538847
A-Qd / Leo IV / Plummer	13.997157	13.997143	13.997022	13.995652	13.983604	13.846925
A-Qd / Leo IV / cusp	13.878786	13.878726	13.878197	13.872468	13.828534	13.462055
A-Qd / Pegasus III / Plummer	19.361049	19.361020	19.360765	19.357862	19.332365	19.044348
A-Qd / Pegasus III / cusp	19.205110	19.204994	19.203988	19.193043	19.109255	18.448160
A-Qd / Hydrus I / Plummer	3.937669	3.937660	3.937584	3.936714	3.929065	3.842062
A-Qd / Hydrus I / cusp	3.489304	3.489260	3.488873	3.484526	3.447946	3.079056

BASE default values are reproduced within the registered gate; their full grids are archived. The negative values occur only in the A-f1/2 Hydrus I 10 pc column. All Q-variant controls below are positive.

Object / profile	BASE control	A-Qs control	A-Qd control
Leo IV / Plummer	14.014605	14.488396	13.997158
Leo IV / cusp	13.896449	14.369924	13.878792
Pegasus III / Plummer	19.373652	19.854737	19.361052
Pegasus III / cusp	19.217916	19.698709	19.205121
Hydrus I / Plummer	3.994097	4.409681	3.937670
Hydrus I / cusp	3.556883	3.953465	3.489309

Appendix B. Softened-profile virial matrix

Refined V/V_N , baseline f and Q . All 72 GQUMOND values and 12 paired controls are positive on every grid. Each profile preserves total mass and projected half-light radius.

Object / ϵ/r_h	Control	.03 pc	.1 pc	.3 pc	1 pc	3 pc	10 pc
Leo IV / 0.01	13.89878	13.89877	13.89871	13.89821	13.89274	13.85001	13.48689
Leo IV / 0.03	13.90442	13.90441	13.90436	13.90391	13.89892	13.85895	13.50670
Leo IV / 0.1	13.91608	13.91608	13.91604	13.91571	13.91192	13.87929	13.55813
Leo IV / 0.3	13.92178	13.92178	13.92175	13.92150	13.91868	13.89405	13.62860
Pegasus III / 0.01	19.22531	19.22530	19.22519	19.22423	19.21366	19.13161	18.47515
Pegasus III / 0.03	19.24395	19.24394	19.24384	19.24296	19.23321	19.15650	18.51800
Pegasus III / 0.1	19.27451	19.27450	19.27442	19.27372	19.26587	19.20077	18.61591
Pegasus III / 0.3	19.29534	19.29533	19.29527	19.29475	19.28883	19.23757	18.72142
Hydrus I / 0.01	3.55769	3.55769	3.55765	3.55731	3.55341	3.51953	3.16368
Hydrus I / 0.03	3.56015	3.56015	3.56012	3.55982	3.55651	3.52744	3.20610
Hydrus I / 0.1	3.56359	3.56359	3.56356	3.56335	3.56094	3.53971	3.29957
Hydrus I / 0.3	3.54445	3.54445	3.54443	3.54427	3.54250	3.52688	3.34956

References and record identifiers

[1] W. S. Paik, UDCT v7.10 - Pre-registration: sensitivity protocol, reviewed revision, 5 October 2026. Attached authoritative registration. Original draft 4 October 2026.

[2] W. S. Paik, Positive global virial work with local Jeans obstructions: UDCT v7.09 post-registration results, 4 October 2026. DOI: 10.5281/zenodo.23130843.

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[4] M. Milgrom, Generalizations of quasilinear MOND, Physical Review D 108, 084005 (2023). arXiv:2305.01589. Direct theoretical precedent, as identified in the frozen registration.

[5] W. S. Paik, UDCT v7.08 prior-art correction. DOI: 10.5281/zenodo.23117541. UDCT v7.06 sign audit and frozen dependency: DOI 10.5281/zenodo.23039754.

[6] W. S. Paik, UDCT v7.10 post-registration execution archive, 8 October 2026. results.json and independent_audit.json; accompanying code and provenance record.

Final statement. The registered outcomes are G2/J1/K2 with passed numerical gates. They describe the finite frozen implementation and sampled profiles. They neither validate a complete gravitational theory nor establish a universal failure of MOND.